Computational Statistics and Simulation: Section 3. Bootstrap and Jackknife Resampling Techniques

Master in Mathematics and Applications

Isabel Pereira,
isabel.pereira@ua.pt

Universidade de Aveiro,
2025/26

Section 3: Bootstrap and Jackknife Resampling Techniques

1. Introduction
2. Bootstrap method
3. Jackknife methodology

1. Introduction

In this section we will present two methods that can be used as an alternative to the Monte Carlo methods mentioned above when no statistical model is available.

Resampling Techniques:

- Bootstrap
- Jackknife

which, from a single sample generate **new samples to the image from the original**.

Resampling methods treat an observed sample as a finite population; then samples are generated from the original to estimate population characteristics and make inferences about the population.

2. Bootstrap method

- The Bootstrap Method was introduced in 1979 by Efron. The Bootstrap methods are a **class of Monte Carlo non-parametric methods that estimate the population distribution by resampling**. The term "bootstrap"can be addressed to non-parametric/parametric bootstrap. In this section we address the study of bootstrap method. .
- The finite population distribution represented by the sample can be seen as a **pseudo-population**, with similar characteristics to those of the true population.
- By repeatedly generating random samples from this pseudo-population (resampling), the sampling distribution of a statistic can be estimated.
- bootstrap generates samples randomly from the **empirical distribution of the sample**.
- Properties of an estimator such as **bias or standard deviation can be estimated by resampling**.

Def 2.1: empirical distribution function
Let $\mathbf{x} = (x_1, \ldots, x_n)$be an observed random sample from population with distribution function F_X. The associated distribution function to $X*$ that uniformly assigns

$$P(X* = x_i) = \frac{1}{n}$$

is the empirical distribution function (edf), denoted by F_n

- $F_n(x) \hookrightarrow$ estimator of $F_X(x)$
- in bootstrap we have two approaches:
  $F \longrightarrow X \longrightarrow F_n$
  $F_n \longrightarrow X* \longrightarrow F*_n$

2. Bootstrap method

Resampling by Bootstrap

We pretend to estimate the parameter θ through the estimator $\hat{\theta}$ using bootstrap sampling: We have to generate bootstrap samples by resampling from the sample $x = (x_1, \ldots, x_n)$

BOOT algorithm

- for each bootstrap replicate, indexed in $b = 1, 2, \ldots, B$:
  (a) generate the bootstrap sample $x^{*(b)} = (x_1^*, \ldots, x_n^*)$, using **sampling with replacement** from the observed sample $(x_1, \ldots, x_n)$
  (b) Calculate the bth replicate, $\hat{\theta}^{(b)}$ using the bootstrap sample $x^{*(b)}$
- The bootstrap estimate of $F_{\hat{\theta}}$ is the empirical distribution function of the replicates $\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(B)}$, given by:

$$F_n^*(x) = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}_{\{\hat{\theta}^{(b)} \leqslant x\}}$$

Example 2.1: empirical df F_n and boostrat sample
Let $\mathbf{x} = (x_1, \ldots, x_{10}) = (2, 2, 1, 1, 5, 4, 4, 3, 1, 2)$
How can we proceed to get a bootstrap sample from x?

We randomly select, with replacement, the values 1,2,3,4 or 5 with probabilities 0.3,0.3,0.1,0.2 or 0.1, respectively. Then the df $F_{X*}(x)$ of the bootstrap sample coincides with $F_n(x)$:

$$F_{X*}(x) = F_n(x) = \begin{cases} 0, & x < 1 \\ 0.3, & 1 \leq x < 2 \\ 0.6, & 2 \leq x < 3 \\ 0.7, & 3 \leq x < 4 \\ 0.9, & 4 \leq x < 5 \\ 1, & x \geq 5\,. \end{cases}$$

Very Important: If F_n is not close enough to F_X, then the distribution of the replicates is not close to F_X, as well.
Assume that $X \sim P(2)$; resampling from x using large B produces good estimate to F_n; however is not a good estimate of F_X; why?

Bias bootstrap estimation

Recall that the bias of the estimator $\hat{\theta}$ of θ is defined:

$$bias(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Bias bootstrap estimation uses bootstrap replicates of $\hat{\theta}$ to estimate the sampling distribution of $\hat{\theta}$

Def 2.2: bootstrap estimate of the bias

$$\hat{bias}^*(\hat{\theta}) = \bar{\hat{\theta}}^* - \hat{\theta},$$

with

$$\bar{\hat{\theta}}^* = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}^{(b)}$$

and

$$\hat{\theta} = \hat{\theta}(\mathbf{x}) = \hat{\theta}(x_1, \ldots, x_n)$$

Example 2.2: bootstrap estimate of the bias of correlation coefficient.

The database law in the bootstrap library is from Efron and Tibshirani. The data.frame contains data referring to Law School Average Test(LSAT) and Grade-Point Average(GPA) for 15 Law Faculties.

LSAT	576	635	558	578	666	580	555	661	651	605	653	575	545	572	594
GPA	339	330	281	303	344	307	300	343	336	313	312	274	276	288	296

Calculate the bootstrap estimate of the bias of the sample correlation coefficient.

This database is actually a random sample of the universe of 82 Law schools in law82.

2. Bootstrap method

```
 library(bootstrap)
 data(law)
# Estimativa Bootstrap do viés do
# coef Correlação R

# estimativa para a amostra law de dimensão n=15
theta.hat <- cor(law$LSAT, law$GPA)

# estimativa bootstrap do viés
B <- 2000  # número de réplicas bootstrap
n <- nrow(law)
theta.b <- numeric(B)
    for (b in 1:B) {
        i <- sample(1:n, size = n, replace = TRUE)  # i é o vector dos índices
        LSAT <- law$LSAT[i]
        GPA <- law$GPA[i]
        theta.b[b] <- cor(LSAT, GPA)
    }
    bias <- mean(theta.b) - theta.hat
    bias

>   bias
[1] 0.007822616
```

Standard deviation bootstrap estimation

The bootstrap estimate of the standard deviation of an estimator $\hat{\theta}$ is the empirical standard deviation of bootstrap replicates $\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(B)}$

Def 2.3: bootstrap estimate of standard deviation

$$\hat{se}^*(\hat{\theta}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{\theta}^{(b)} - \bar{\hat{\theta}}^*)^2},$$

with

$$\bar{\hat{\theta}}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}^{(b)}$$

Example 2.3: bootstrap estimate of the sd of correl. coef.

Returning to Example 2.2, calculate the bootstrap estimate of the standard deviation of the sample correlation coefficient.

It should be noted that in this example we have access to the entire population in the law82, so it is possible to compare the estimates based on the sample law with the population value of the coefficient of correlation.

```
>     print(cor(law$LSAT, law$GPA)) # coef correlação empírico
[1] 0.7763745
>     print(cor(law82$LSAT, law82$GPA)) # coef correlação populacional
[1] 0.7599979
```

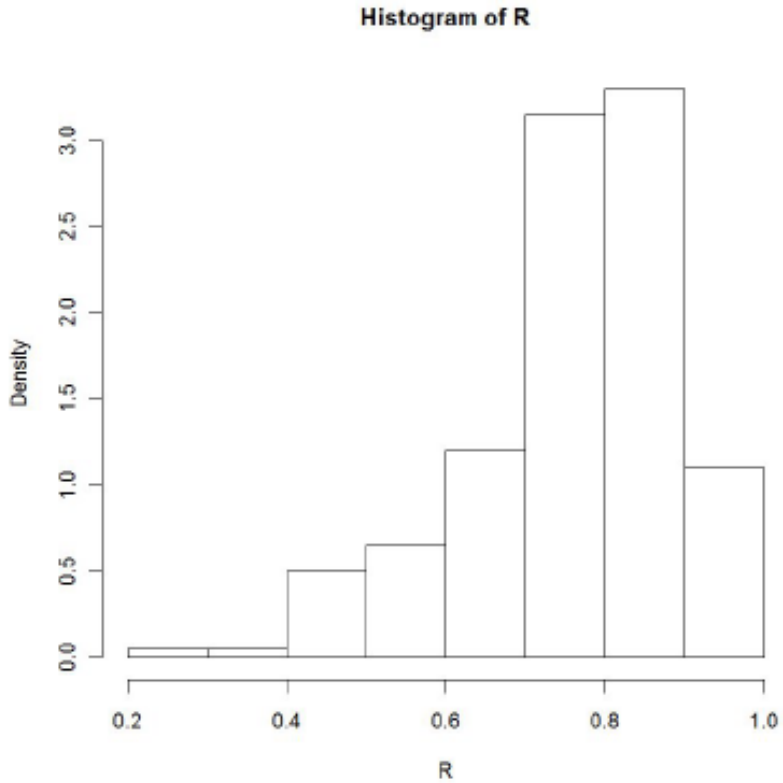
2. Bootstrap method

```
# Estimativa Bootstrap do desvio padrão do
# coef Correlação R

B <- 200          #número de réplicas
n <- nrow(law)    #dimensão da amostra
R <- numeric(B)   #armazena as réplicas de R
```

```
#Estimativa Bootstrap do desvio padrão de R
for (b in 1:B) {
    # selecção dos índices aleatoriamente
    i <- sample(1:n, size = n, replace = TRUE)
    LSAT <- law$LSAT[i]      # i é o vector dos índices
    GPA <- law$GPA[i]
    R[b] <- cor(LSAT, GPA)
 }
 #output

 >     print(se.R <- sd(R))
[1] 0.1297349
>     hist(R, prob = TRUE)
```



Results analysis

- Then the bootstrap estimate of the standard deviation of the sample correlation coefficient R, $se(R) = se(\hat{\rho}) = se(\hat{\theta})$, is given by $\hat{se}(\hat{\theta}^*) = 0.1297349$;

- in this case we have access to the exact value of $\rho = 0.7599979$

- so the theoretical value for the standard deviation of R (under normality), for n = 15, is

$$se(\hat{\theta}) = se(R) = \sqrt{\frac{(1-\rho^2)^2}{n}} = \frac{1 - 0.7599979^2}{\sqrt{15}} \approx 0.113$$

3.1 Jackknife methodology

Another form of resampling is the Jackknife methodology, proposed by Quenouille (1949, 1956) to estimate bias and by Tukey (1958) to estimate the standard deviation, (a few decades before the bootstrap)

- Generally speaking, we can say that it is a resampling methodology of an estimator
$$\hat{\theta}_n = \hat{\theta}_n(X_1, \ldots, X_n)$$
- that leaves out an observation in each resampling of size $n-1$
$$X^{(j)} := (X_1, \ldots, X_{j-1}, X_{j+1}, \ldots, X_n), j = 1, \ldots, n$$
called Jackknife samples
- Based on Jackknife samples, we calculate
$$\hat{\theta}^{(j)} := \hat{\theta}_{n-1}(X^{(j)}), j = 1, \ldots, n$$

Def 3.1: Jackknife estimate for the bias of $\hat{\theta}_n$

$$\widehat{bias}_{jack}(\hat{\theta}_n) = (n-1)(\overline{\hat{\theta}^{(\cdot)}} - \hat{\theta}_n)$$

with

$$\overline{\hat{\theta}^{(\cdot)}} = \frac{1}{n}\sum_{j=1}^{n} \hat{\theta}^{(j)}$$

The form of this estimator is motivated by the following result:

Lemma 3.1

Let $X_1, \ldots, X_n$ i.i.d. r.v.'s; suppose that

$$bias(\hat{\theta}_n) = \frac{a}{n} + \frac{b}{n^2} + O(\frac{1}{n^3}), \forall n \in \mathbf{N}, a \neq 0.$$

Then

$$E[\widehat{bias}_{jack}(\hat{\theta}_n)] = bias(\hat{\theta}_n) + O(\frac{1}{n^2})$$

3. Jackknife methodology

In R programme, to leave out one of the components of a vector it is simply done

```
x <- 1:5
for (i in 1:5) {
print(x[-i])
}
[1] 2 3 4 5
[1] 1 3 4 5
[1] 1 2 4 5
[1] 1 2 3 5
[1] 1 2 3 4
```

Example 3.1: Jackknife estimator for the bias of Ratio
The data patch (bootstrap)of Efron and Tibshirani contains measurements of a certain hormone in the bloodstream of eight individuals after using a drug. The parameter of interest is

$$\theta = \frac{E[new] - E[old]}{E[old] - E[placebo]}$$

If $|\theta| \leqslant 0.2$, means that old and new drugs are bioequivalents. The statistics of interest is the Ratio:

$$\hat{\theta}_n := \frac{\bar{Y}}{\bar{Z}}$$

where Y:= new-old and Z:= old-placebo

Calculate Jackknife estimate of the bias of the Bioequivalence Ratio statistics, $\hat{\theta}_n$.

3. Jackknife methodology

Jackknife Estimator of the bias of Ratio statistics

```
data(patch, package = "bootstrap")
> patch
  subject placebo oldpatch newpatch     z     y
1       1    9243    17649    16449  8406 -1200
2       2    9671    12013    14614  2342  2601
3       3   11792    19979    17274  8187 -2705
4       4   13357    21816    23798  8459  1982
5       5    9055    13850    12560  4795 -1290
6       6    6290     9806    10157  3516   351
7       7   12412    17208    16570  4796  -638
8       8   18806    29044    26325 10238 -2719

n <- nrow(patch)
y <- patch$y
z <- patch$z
theta.hat <- mean(y) / mean(z)
> print (theta.hat)
[1] -0.0713061

# Calcular as réplicas Jackknife,
# deixando uma observação de fora
theta.jack <- numeric(n)
 for (i in 1:n){
   theta.jack[i] <- mean(y[-i]) / mean(z[-i])
   }
 bias <- (n - 1) * (mean(theta.jack) - theta.hat)

> print(bias)  # estimativa jackknife do viés
[1] 0.008002488
```

Def 3.2: Jackkife estimate for the standard deviation of $\hat{\theta}_n$

$$\widehat{se}_{jack}(\hat{\theta}_n) = \sqrt{\frac{n-1}{n}\sum_{j=1}^{n}(\hat{\theta}^{(j)} - \overline{\hat{\theta}^{(\cdot)}})^2}$$

with

$$\overline{\hat{\theta}^{(\cdot)}} = \frac{1}{n}\sum_{j=1}^{n}\hat{\theta}^{(j)}$$

Example 3.2: Jackknife estimator for the sd of Ratio estimator

Calculate the estimate of standard deviation for the Ratio estimator, using data if Example 3.1

In R programme:

```
se <- sqrt((n-1) *  mean((theta.jack - mean(theta.jack))^2))
> print(se)
[1] 0.1055278
```

Conclusion: Exists bioequivalence between new and old drugs.

References - Bibliography

- Quenouille, M. (1949). Approximate tests of correlation in time series.*Journal of the Royal Statistical Society*, Series B, 11, 68-84.
- Quenouille, M. (1956). Notes on bias in estimation. *Biometrika*, 43,353-360.
- Tukey, J. W. (1958). Bias and confidence in not quite large samples (abstract). *The Annals of Mathematical Statistics*, 29, 614.

In R programme are available the following packages:

- Library bootstrap: collection of functions to data, for Efron, B. and Tibshirani, R.J., An Introduction to the Bootstrap (Chapman & Hall/CRC Monographs on Statistics & Applied Probability), 1993.
- Library boot: collection of functions to data, for Davison, A.C. and D. V. Hinkley, D.V., Bootstrap Methods and their Application (Cambridge Series in Statistical and Probabilistic Mathematics), 1997.